

已知 $f_X(x) = \begin{cases} \frac{2x}{\pi^2}, & 0 < x < \pi \\ 0, & \text{其他} \end{cases}$

求 $Y = \sin X$ 的分布密度.

解: Y 的分布函数为 $F_Y(y)$. 则

当 $y \leq 0$ 时, $F_Y(y) = 0$, 当 $y \geq 1$ 时, $F_Y(y) = 1$.

当 $0 < y < 1$ 时, 有

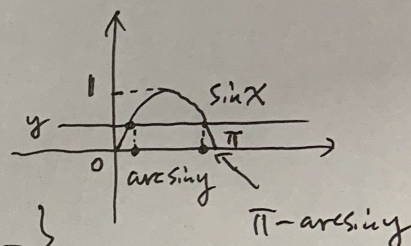
$$F_Y(y) = P\{Y \leq y\} = P\{\sin X \leq y\}$$

$$= P\{0 \leq X \leq \arcsin y\} + P\{\pi - \arcsin y \leq X \leq \pi\}$$

$$= \int_0^{\arcsin y} \frac{2x}{\pi^2} dx + \int_{\pi - \arcsin y}^{\pi} \frac{2x}{\pi^2} dx$$

$$= \frac{1}{\pi^2} (\arcsin y)^2 + \frac{1}{\pi^2} [\pi^2 - (\pi - \arcsin y)^2]$$

$$= \frac{1}{\pi^2} \cdot 2\pi \arcsin y = \frac{2}{\pi} \arcsin y$$



从而有 $F_Y(y) = \begin{cases} 0, & y \leq 0 \\ \frac{2}{\pi} \arcsin y, & 0 < y < 1 \\ 1, & y \geq 1 \end{cases}$

于是 $Y = \sin X$ 的分布密度为

$$f_Y(y) = \begin{cases} \frac{2}{\pi \sqrt{1-y^2}}, & 0 < y < 1 \\ 0, & \text{其他} \end{cases}$$